```
        super(MultiHeadRelationalModule, self).__init__()
        self.conv1_ch = 16
        self.conv2_ch = 20
        self.conv3_ch = 24
        self.conv4_ch = 30
        self.H = 28
        self.W = 28
        self.node_size = 64
        self.lin_hid = 100
        self.out_dim = 5
        self.ch_in = 3
        self.sp_coord_dim = 2
        self.N = int(7**2)
        self.n_heads = 3

        self.conv1 =
     nn.Conv2d(self.ch_in,self.conv1_ch,kernel_size=
(1,1),padding=0)     1
        self.conv2 =
     nn.Conv2d(self.conv1_ch,self.conv2_ch,kernel_size=
(1,1),padding=0)
        self.proj_shape =
(self.conv2_ch+self.sp_coord_dim,self.n_heads *
     self.node_size)
        self.k_proj = nn.Linear(*self.proj_shape)
        self.q_proj = nn.Linear(*self.proj_shape)
        self.v_proj = nn.Linear(*self.proj_shape)

        self.k_lin = nn.Linear(self.node_size,self.N)
2
        self.q_lin = nn.Linear(self.node_size,self.N)
        self.a_lin = nn.Linear(self.N,self.N)

        self.node_shape = (self.n_heads, self.N,self.node_size)
        self.k_norm = nn.LayerNorm(self.node_shape,
elementwise_affine=True)
        self.q_norm = nn.LayerNorm(self.node_shape,
elementwise_affine=True)
        self.v_norm = nn.LayerNorm(self.node_shape,
elementwise_affine=True)

        self.linear1 = nn.Linear(self.n_heads * self.node_size,
self.node_size)
        self.norm1 = nn.LayerNorm([self.N,self.node_size],
            elementwise_affine=False)
        self.linear2 = nn.Linear(self.node_size, self.out_dim)

    def forward(self,x):
        N, Cin, H, W = x.shape
        x = self.conv1(x)
```